

TezHack 2026

Waste Material Classification for Better Segregation

Detailed Research-Oriented Work Plan, Implementation Blueprint & Rollout Roadmap

Proposed product: AI-Assisted Waste Segregation & Safe Disposal Guidance System

Purpose of this document

This report converts the TezHack problem statement into a concrete 48-hour execution plan and a technically defensible path toward real-world deployment. It covers the complete plan: taxonomy, datasets, ML experiments, uncertainty handling, safety, policy mapping, UI, evaluation, team allocation, deployment and future rollout.

MASTER PRINCIPLE

Train the model to recognize materials, measure when it fails, use uncertainty to refuse unreliable predictions, and use a separate policy layer to turn predictions into safe, locally adaptable disposal guidance.

1. Executive Summary

The MVP will accept a waste photograph, perform image-quality checks, classify a supported waste material using a lightweight pretrained vision model, estimate prediction reliability, and map the result to an action-oriented disposal category. Hazardous and special materials will receive conservative guidance rather than ordinary-bin instructions.

The system is deliberately designed as a hybrid rather than a pure classifier:

Computer Vision: What does this image appear to contain?
Uncertainty Layer: Is the prediction reliable enough to show?
Policy Engine: What disposal stream is appropriate?
Location Layer: What does the relevant local system recommend?

2. Project Goals

Primary goal: Build an image-based decision-support prototype that helps users segregate common household waste.

- Use a public labelled waste-image dataset.
- Perform documented preprocessing and augmentation.
- Fine-tune a pretrained lightweight vision model.
- Report accuracy, precision/recall, Macro-F1 and a confusion matrix.
- Display prediction confidence.
- Return Uncertain for unclear, unsupported or low-confidence images.
- Provide rule-based disposal guidance.
- Separate hazardous/e-waste/sanitary pathways from ordinary waste.
- Include mandatory safety and local-guidance disclaimers.
- Design the MVP so it can evolve into a location-aware civic platform.

3. Research Questions & Hypotheses

Research question	How we will test it
RQ1. Can a lightweight pretrained CNN classify waste effectively?Net-B0 and report class-wise and aggregate metrics.	Transfer learning
RQ2. Does real-world data improve generalization?Are TrashNet-only, RealWaste-only and combined training; test on held-out and external images.	Controlled vs. real-world
RQ3. Can uncertainty reduce unsafe overconfidence?Can we reject primary classification against confidence/entropy-based rejection.	Confidence/entropy
RQ4. Can disposal rules be separated from recognition?Can we add a policy layer whose rules can be changed without retraining the vision model.	ML model + policy

- H1:** Transfer learning will provide a strong baseline under limited compute and time.
- H2:** Combining controlled and real-world imagery should improve robustness to user photographs.
- H3:** Rejection of low-confidence/high-entropy predictions should reduce confident errors, at the cost of lower coverage.
- H4:** Separating recognition from policy enables local rule updates without model retraining.

4. Final Waste Taxonomy

The project will maintain two taxonomies. The user-facing taxonomy is action-oriented; the ML taxonomy is material/object-oriented. This prevents the model from making policy claims that cannot be inferred from pixels alone.

ID	User-facing category	Examples / role
C1	Organic / Wet Waste	Food scraps, fruit/vegetable peels, biodegradable household waste.
C2	Dry Recyclable Waste	Paper, cardboard, suitable plastic, glass and metal; subject to local acceptance.
C3	General / Residual Waste	Ordinary non-recoverable or contaminated residual material.
C4	E-Waste	Phones, chargers, keyboards, cables and electrical/electronic items.
C5	Domestic Hazardous / Special Waste	Batteries, chemicals and other materials needing special handling.
C6	Sanitary Waste	Diapers, sanitary products and similar contaminated hygiene waste.
C7	Uncertain	Safety state for insufficient evidence, poor image quality, unsupported/OOD input.

Internal ML labels: Food/Organic, Paper, Cardboard, Plastic, Glass, Metal, Textile, E-Waste/Electronics, Battery/Hazardous, Sanitary, Residual/Other.

Important: “Recyclable” is not treated as a universal visual truth. Actual acceptance can depend on composition, contamination, infrastructure and local policy.

5. Dataset Strategy

Dataset	Scale / structure	Role	Main limitation
TrashNet	2,527 images; 6 classes	Fast, labelled classification baseline.	Controlled imagery and imbalance; trash class is much smaller than others.
RealWaste	≈4,752 images; 9 material types	Real-world/landfill variability and stronger robustness.	Less systematic than TrashNet and needs harmonization.
TACO	≈1,500 images / 4,784 annotations	Enterprise-level robustness test; litter-in-context.	Primarily detection/segmentation-oriented rather than classification.
WhichWaste (optional)	≈11,270 images; 24 categories	Future expansion / additional data source.	Too many categories for the 48-hour MVP.

Dataset links

TrashNet: <https://github.com/garythung/trashnet>
RealWaste: <https://archive.ics.uci.edu/dataset/908/realwaste>
TACO: <https://tacodataset.org/>
WhichWaste: <https://doi.org/10.7910/DVN/ROFOT5>

Research design: TrashNet is the controlled baseline; RealWaste adds real-world variation; TACO is kept outside training and used as an external robustness test wherever label compatibility permits.

6. Data Engineering & Integrity

- Inspect every class and calculate image counts and class percentages.
- Check for corrupted files and obvious duplicates.
- Create a harmonized label map across datasets.
- Split into train/validation/test without leakage.
- Keep the external stress-test set isolated from training.
- Document dataset version, source, class mapping and split seed.
- Use class weighting or balanced sampling only where analysis shows it is necessary.

Data leakage prevention is mandatory. A model evaluated on images or near-duplicates it has effectively seen during training cannot support a credible generalization claim.

7. ML Architecture

Primary backbone: EfficientNet-B0 pretrained on ImageNet.

Pipeline: Image → 224×224 preprocessing → ImageNet-normalized tensor → EfficientNet-B0 → dropout/classification head → ML class probabilities.

Why this choice: it is lightweight enough for rapid fine-tuning on a Colab/Kaggle GPU, benefits from transfer learning, and is substantially easier to operationalize within the 48-hour deadline than training a large network from scratch.

Fallback: MobileNetV3-Large if EfficientNet-B0 creates integration or inference issues. Do not let architecture experimentation consume the core build window.

8. Training Plan

Setting	Initial choice
Input	224 × 224
Weights	ImageNet pretrained
Optimizer	AdamW
Initial fine-tuning LR	≈1e-4
Weight decay	≈1e-4
Batch size	32, adjusted to available GPU memory
AMP	Enabled
Epochs	10–20 with early stopping
Loss	Cross-entropy; class weighting if justified by imbalance
Seed	Fixed for reproducibility

Training will proceed in two stages: first train the classification head with most of the backbone frozen; then unfreeze selected upper layers for fine-tuning. Record hardware, training time, hyperparameters and checkpoint version.

9. Image Preprocessing & Augmentation

Training: RandomResizedCrop, horizontal flip, small rotation, moderate ColorJitter and ImageNet normalization.

Validation/Test: deterministic resize/crop and normalization only; no random augmentation.

The objective is not merely mathematical regularization. Augmentations should mimic real phone-camera variation such as angle, scale and lighting changes while avoiding unrealistic transformations.

10. Experimental Matrix

Experiment	Training data	Purpose
A	TrashNet	Controlled baseline.
B	RealWaste	Real-world dataset baseline.
C	TrashNet + RealWaste	Test whether combined training improves generalization.
D	Combined + uncertainty	Measure the safety/coverage trade-off created by rejection.
E	External TACO/real-world images	Stress-test performance outside the main training distribution.

11. Evaluation Framework

Primary: Accuracy, Macro Precision, Macro Recall and Macro F1.

Secondary: Per-class precision/recall/F1, confusion matrix, confidence distribution, uncertain rate and external-test performance.

Macro-F1 is critical because overall accuracy can hide poor performance on minority classes. Per-class recall is particularly important for E-Waste, Hazardous/Special and Sanitary pathways.

Application-specific safety metric: Unsafe Routing Error Rate = special/hazardous samples that are incorrectly routed toward an ordinary/general/recyclable pathway, divided by the relevant special-item evaluation set. The exact operational definition will be fixed before final evaluation.

12. Confusion Matrix & Prediction Summary

The evaluation dashboard will show the full confusion matrix and a prediction-level table containing actual class, predicted class, confidence, uncertainty status and correctness.

Image	Actual	Prediction	Confidence	Correct
001	Plastic	Plastic	94%	Yes
002	Paper	Cardboard	63%	No
003	Glass	Glass	96%	Yes

Error analysis will explicitly inspect ordinary confusions such as paper/cardboard and plastic/residual, while separately highlighting safety-relevant errors such as battery→residual or e-waste→residual.

13. Uncertainty & Unclear-Image Handling

The model must be allowed to say “Uncertain.” A softmax classifier can be highly confident on an image that is actually outside its training distribution, so top-1 confidence alone is not treated as a guarantee.

Signals: maximum softmax probability plus predictive entropy. Entropy is calculated as $H(p) = -\sum p_i \log p_i$.

Initial decision: poor image quality → Uncertain; confidence below a validation-calibrated threshold → Uncertain; entropy above a validation-calibrated threshold → Uncertain; otherwise accept the prediction.

Thresholds will be selected empirically on validation data. We will not claim that an arbitrary 0.5/0.7/0.9 threshold is optimal without testing.

14. Image Quality Gate

- Blur: variance-of-Laplacian heuristic.

- Very dark image: mean luminance threshold.
- Very low resolution: minimum image dimensions.
- Multiple objects: MVP asks the user to photograph one item at a time.
- Future production upgrade: object detection for multi-item scenes.

OOD test images will include unsupported objects and non-waste scenes. The desired behavior is not a forced label; it is an Uncertain response.

15. Safety-First Decision Hierarchy

Image → quality check → model → confidence/entropy → special-risk check → policy engine → location-aware guidance → final response.

For potentially hazardous or special materials, the system must not provide a normal-bin recommendation merely because the classifier is confident. It should route to special handling/local-authority guidance.

16. Rule-Based Disposal Engine

The ML model does not directly determine disposal instructions. It outputs a material/object label; a separate rule engine maps that label to a user-facing category and instruction.

Example: Plastic bottle → Plastic → Dry Recyclable pathway → “Follow local acceptance rules; empty/clean where appropriate.”

Rules should be configuration-driven rather than hard-coded. Each rule should support fields such as location, ML class, user category, disposal stream, instruction, warning, source, verification date and version.

17. Location-Aware Future Architecture

For the hackathon, use a local JSON/configuration file. For production, store municipal rules in PostgreSQL and resolve guidance using material + condition + location.

Rule field	Purpose
location / municipality	Determines which local policy applies.
ML class	Connects visual prediction to policy.
user category	Human-readable segregation stream.
disposal stream	Operational action.
instruction	Practical preparation/handling guidance.
warning	Safety-specific message.
source	Authority/verified information source.
verified_at / version	Auditability and policy updates.

18. Mandatory Safety & Disclaimer Policy

Safety & Disposal Notice: This service provides AI-assisted waste classification for segregation guidance. It does not certify hazardous waste, guarantee recyclability, or replace instructions from local municipal authorities, authorized waste collectors, or hazardous-waste handlers. Waste requirements vary by location and by the condition and composition of an item. For hazardous, electronic, medical, chemical, sharp, or otherwise special waste, follow authorized local disposal guidance.

For hazardous/special predictions, the UI should explicitly say that the AI does not certify safe disposal and should direct the user toward local or authorized channels.

19. 48-Hour Execution Schedule

Time	Work	Deliverable
0–2 h	Repository, environment, datasets, experiment tracking.	Working project structure.
2–5 h	Dataset inspection, counts, cleaning, label harmonization.	Dataset report + class chart.
5–8 h	EfficientNet-B0 baseline.	First working checkpoint.
8–12 h	Augmentation + fine-tuning + evaluation.	Metrics + confusion matrix.
12–15 h	TrashNet/RealWaste/combined comparison.	Experiment comparison table.
15–18 h	Confidence + entropy + threshold search + OOD tests.	Uncertainty module + analysis.
18–22 h	Rule engine + safety logic.	Disposal mapping.
22–28 h	Streamlit interface.	Working end-to-end demo.
28–32 h	Evaluation dashboard.	Confusion matrix + prediction summary.
32–36 h	Real-world/TACO/team-photo testing.	Failure set + external results.
36–40 h	Error analysis and targeted fixes.	Final model candidate.
40–44 h	Deployment + complete scenario testing.	Stable live demo.
44–48 h	Presentation, README, architecture and final report.	Submission package.

20. Team Work Allocation

Track	Responsibilities
A — Data & Research	Datasets, cleaning, class statistics, label mapping, experiment log, evaluation report.
B — ML	EfficientNet-B0, training, augmentation, fine-tuning, uncertainty, OOD testing, confusion matrix.
C — Product/UI	Streamlit, upload/camera, results, confidence, uncertain state, disposal cards, safety notice.
D — Integration & Presentation	Rule engine, deployment, testing, architecture diagrams, demo script, slides.

21. MVP Application Flow

Home → Upload/Capture → Preview → Analyze → Result.

Result screen should contain: detected material, user-facing category, confidence status, recommended action, explanation/why, preparation guidance where appropriate, and the safety/local-rule disclaimer.

Uncertain screen: “The system could not determine the waste category reliably. The image may be blurry, contain multiple items, or be outside the supported categories. Please upload a clear image showing one waste item.”

22. Final Results Table

Model/Data	Accuracy	Macro F1	External F1	Uncertain %	Safety Error Rate
TrashNet + EfficientNet-B0	TBD	TBD	TBD	TBD	TBD
RealWaste + EfficientNet-B0	TBD	TBD	TBD	TBD	TBD
Combined + EfficientNet-B0	TBD	TBD	TBD	TBD	TBD
Combined + uncertainty	TBD	TBD	TBD	TBD	TBD

All numerical results must come from the actual experiments. No accuracy or confidence value should be invented for the final presentation.

23. Production Rollout Roadmap

Phase	Scope	Objective
1 — TezHack MVP	Streamlit + EfficientNet-B0 + rule config.	Demonstrate feasibility and research evidence.
2 — Campus pilot	Deploy to a controlled university/campus environment.	Collect realistic images and user feedback.
3 — Human-reviewed data	Review incorrect/uncertain samples before adding to training.	Build a local high-quality dataset.
4 — Local pilot	Partner with a municipality/waste operator.	Validate real disposal guidance.
5 — Production platform	Next.js/React + FastAPI + PostgreSQL + ML inference service.	Scalable application.
6 — Multi-object detection	Detect multiple waste objects in a scene.	Handle real household/bin photographs.
7 — Collection-point integration	Authorized e-waste/special-waste locations.	Give users an actionable next step.
8 — Municipal analytics	Privacy-preserving aggregated insights.	Support awareness and waste-management decisions.

24. Production Architecture

Client (web/mobile) → API Gateway → Vision Service + Policy Engine + Location Service → PostgreSQL → Recommendation/Safety Layer → User.

The production database should maintain locations, municipalities, disposal rules, collection points, waste classes, model versions and reviewed feedback. Model versions should be auditable and rollback-capable.

25. Human Feedback & Model Improvement

The UI should include a lightweight feedback mechanism such as “Correct / Incorrect.” Incorrect predictions become candidate samples, not automatic training data.

Pipeline: Prediction → user feedback → human review → approved label → dataset version → retraining → evaluation → model registry → deployment.

This prevents uncontrolled user-generated data from poisoning the training set.

26. Privacy & Responsible AI

- Collect only information needed for the service.
- Avoid unnecessary retention of uploaded photographs.
- Use consent and clear privacy communication where appropriate.
- Do not automatically train on user images.
- Maintain model/version and policy audibility.
- Communicate model limitations clearly.
- Prefer conservative escalation over unsafe certainty for high-risk materials.

27. Success Criteria

Area	Success condition
Functional	Upload/camera, prediction, confidence, Uncertain, disposal guidance and evaluation views all work.
ML	A trained model beats a simple baseline; Macro-F1 and per-class metrics are reported.
Generalization	External real-world testing is performed and discussed.
Safety	Special/hazardous pathways are separated and low-confidence/OOD inputs can be rejected.

Area	Success condition
Research	Datasets, experiments, preprocessing, results and limitations are reproducible.
Deployment	The application can be run reliably and demonstrated end-to-end.

28. Final Project Positioning

Do not position the project as “a CNN that classifies garbage.”

Position it as:

“AI-Assisted Waste Segregation with Uncertainty-Aware Disposal Guidance.”

The innovation is the combination of material recognition, uncertainty detection, safety-aware routing and a localizable policy layer—not the invention of a new CNN architecture.

This approach satisfies the TezHack brief while creating a credible route from a 48-hour prototype to a controlled campus pilot, then to a municipality-aware waste guidance platform.

29. Final Deliverables Checklist

- Working Streamlit application.
- Trained EfficientNet-B0 checkpoint.
- Preprocessing and training code/notebook.
- Dataset statistics and harmonized label map.
- Confusion matrix and classification report.
- Prediction summary.
- Confidence/entropy uncertainty module.
- OOD/uncertain test results.
- Rule-based disposal configuration.
- Safety disclaimer.
- README and reproducibility instructions.
- Architecture diagram.
- Research results table.
- Final presentation and live-demo script.

END OF PROJECT PLAN

Recommended execution order: **Data** → **Baseline** → **Fine-tuning** → **Evaluation** → **Uncertainty** → **Safety/Rules** → **UI** → **Real-world testing** → **Deployment** → **Presentation**.